

memory value appears on the bus. The signals are not co-ordinated with the system clock at all.

Asynchronous memory works fine in lower-speed memory bus systems, but is not nearly as suitable for use in high-speed memory systems. A newer type of DRAM called ‘synchronous DRAM’, or SDRAM, is synchronised to the system clock. This type of memory is much faster than asynchronous DRAM. It is more suitable to the higher-speed memory systems of the newest PCs.

Double Data Rate SDRAM (DDR SDRAM)

A few years ago, ‘regular’ SDRAM was introduced as a proposed replacement for the older FPM and EDO asynchronous DRAM technologies. In the next couple of years, as system bus speeds increase further, double data rate RAM came up: it is a type of SDRAM in which data is sent on both the rising and falling edges of clock cycles in a data burst.

Single Inline Memory Modules (SIMMs)

The SIMM is the most common memory module format in use in the PC world, largely due to the enormous installed base of PCs that use them; in new PCs, DIMMs are now overtaking SIMMs in popularity. SIMMs are available in 30-pin and 72-pin flavours. 30-pin SIMMs are the older standard, and were popular on third and fourth generation motherboards. 72-pin SIMMs are used on fourth, fifth and sixth generation PCs.

SIMMs are placed into special sockets on the motherboard. The sockets are specifically designed to ensure that once inserted, the SIMM will be held in place tightly. SIMMs are secured into their sockets (in most cases) by inserting them at an angle (usually about 60 degrees from the motherboard) into the base of the socket and then tilting them upward until they are perpendicular to the motherboard. Special metal clips on either side of the socket snap in place when the SIMM is inserted correctly. The SIMM is also keyed with a notch on one side, to make sure it isn’t put in backwards.